LEONARDO FERREYRA CANAVAL

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Professional Summary

Software Engineer with 4+ years of experience designing scalable backend systems, AI-driven architecture, and cloud-native solutions. Specializes in Java/Spring Boot, Python, RESTful APIs, and distributed systems, with deep expertise in AI agent frameworks (CrewAI, LangGraph), LLM observability (Phoenix/Arize), and MCP-based automation. At Kinara Systems, owned end-to-end design and delivery of multi-agent AI pipelines for real-time network analytics — integrating self-hosted and cloud LLMs, building LLM proxy infrastructure, and driving traceability with Phoenix. At Enviame, improved system reliability through Redis-based batching and automated developer workflows via MCP integrations with JIRA, Confluence, and Bitbucket. Proficient in Node.js, Python, Docker, Kubernetes, Redis, AWS, and GCP. Passionate about building intelligent, observable, and production-ready AI systems.

Skills

AI & Agents: CrewAI, LangGraph, MCP (Model Context Protocol), RAG Pipelines, LLM Orchestration, Phoenix/Arize (LLM Tracing & Observability)

Backend: Java / Spring Boot, Python (Flask, Django), Node.js (NestJS, Express), RESTful APIs, JWT Auth, Redis, MQTT

Frontend: React / Redux, Angular, Vue.js, Next.js, Flutter, TypeScript

Cloud & DevOps: GCP (Cloud Run, Cloud Functions, Firestore), AWS, Docker, Kubernetes, CI/CD, Bitbucket Pipelines

Databases: MySQL, PostgreSQL, InfluxDB, Firestore, Redis

Methodologies: Domain-Driven Design, MVC, OOP, Agile / Scrum, Clean Architecture

Languages: Spanish (Native), English (Professional Working — B2 TOEFL)

Work Experience

Software Developer | **Kinara Systems** | Canada (Remote) *11/2025 – 04/2026*

- Owned end-to-end design and delivery of a multi-agent AI pipeline for real-time Wi-Fi network analytics, leveraging CrewAI and LangGraph to orchestrate specialized agents for telemetry analysis, log classification, ping-pong detection, and sticky-client diagnosis.
- Architected and implemented an LLM proxy layer supporting dynamic routing between self-hosted models (Qwen, DeepSeek, ZGLM) and cloud providers (OpenAI, Anthropic), enabling seamless model swapping and A/B evaluation across the system.
- Integrated Phoenix (Arize) for end-to-end LLM observability and tracing — capturing MCPClient calls, CrewAI agent sessions, and tool-call execution to identify latency bottlenecks, hallucination patterns, and model inconsistencies across providers.
- Designed and owned the Streaming Telemetry Analyzer and Streaming Log Analyzer services — Python backends that consume MQTT-based network data, apply LLM-driven classification, and expose runtime-configurable APIs for dynamic threshold and pattern updates without service restarts.
- Built configuration management APIs for both analyzers with YAML-based persistence, enabling runtime updates to LLM parameters, alert thresholds, and monitoring patterns — validated via full integration test suites.
- Evaluated and benchmarked multiple LLMs (GPT-4o, GPT-OSS-120B, DeepSeek V3, Qwen3 235B, ZGLM 4.6) on real network queries, documenting tool-call behavior, latency profiles, and output quality to drive model selection decisions.
- Implemented a TTS (Text-to-Speech) proxy integration across both analyzers, including environment variable propagation, header handling, license-gated activation, and metrics formatting — coordinating alignment across backend services and UI.
- Developed scale-testing infrastructure using InfluxDB and Mosquitto (Docker), generating synthetic datasets of up to 100 APs with dummy client telemetry to stress-test agent pipelines and validate detection logic under load.
- Built the sauble-microagents configuration module and refactored the service to consume client state from the telemetry service, decoupling detector responsibilities and improving system maintainability.
- Led license enforcement implementation across the AI pipeline — integrating claim-token validation, license-gated service termination, and end-to-end testing against deployed license APIs.
- Collaborated with cross-functional teams via Confluence documentation, Jira sprint planning, and code reviews — delivering production-grade deployments with thorough regression testing and root-cause analysis workflows.

Software Developer | **Enviame** | Chile (Remote) *04/2025 – 11/2025*

- Led end-to-end implementation of backend features on Google Cloud (Cloud Run, Cloud Functions, Firestore, Cloud Storage), owning architecture decisions and delivery.
- Designed and optimized MySQL data access patterns, resolving critical performance issues caused by inconsistent data types and inefficient queries, resulting in significantly faster response times.
- Improved system reliability by implementing a Redis-based batching system to regulate outbound requests, reducing rate-limit errors (HTTP 429) and stabilizing external API communication.
- Owned the automation of the internal developer workflow by integrating MCP with JIRA, Confluence, and Bitbucket — including API configuration, workflow orchestration, and automatic PR generation, reducing manual work and improving team productivity.
- Built and maintained backend services using Python (Flask), focusing on clean architecture, maintainability, and performance.

Software Developer | **CIVA** | Lima, Peru *04/2024 – 04/2025*

- Developed backend for the FLIT transportation ERP using Java/Spring Boot, implementing JWT authentication and RESTful APIs.
- Optimized response times by implementing Data Transfer Objects (DTOs), achieving a 30% reduction in latency.
- Migrated from basic authentication to JWT with Spring Boot Security, enabling integration with Microsoft Azure SSO.
- Developed frontend components with React/Redux following Domain-Driven Design principles within Agile/Scrum workflows.

Software Developer | **SOLYMAN Consultores de TI** | Lima, Peru (Remote) *09/2023 – 02/2024*

- Developed a currency and cryptocurrency exchange platform in Odoo using MVC and OOP with real-time REST API integrations.
- Implemented dynamic JavaScript visualizations for trend analysis and a PostgreSQL-backed reporting module for SBS.

Software Engineer | **VALENTOR Producciones S.A.C.** | Lima, Peru (Remote) *04/2022 – 09/2023*

- Architected and delivered 5+ integrated systems for nightclub operations: personnel management (Django + React/TypeScript on AWS), cross-platform mobile app (Flutter + NestJS), POS system (Next.js + Spring Boot), attendance control (Vue.js + .NET + Azure AD with facial recognition), and event marketing platform (Angular + Express.js + Python/Pandas).
- Implemented cloud infrastructure on AWS and GCP with CI/CD, Docker, and Kubernetes — improving deployment speed by 60% and reducing downtime significantly.

Education

B.Sc. Software Engineering | **Universidad Peruana de Ciencias Aplicadas** | Lima, Peru *Expected 06/2026*

GPA: 17.70/20 • Consistently ranked top 5 from 1st–9th semester

Notable Projects: Full-stack Angular + Spring Boot (open source); Vue.js + .NET web application

English Language Studies (B2 TOEFL) | **Instituto Cultural Peruano Norteamericano** | Lima, Peru *03/2022*

Certifications

- Generative AI Fundamentals Specialization — IBM (2025)
- AI Agents — Platzi (2025)
- NestJS — DevTalles (2025)
- Next.js — DevTalles (2025)
- Full Stack MERN Development — Udemy (2024)
- Node.js — DevTalles (2024)
- Microsoft Azure — Microsoft (2023)